SIMATS ENGINEERING
ASSESSMENT TOOL 2

COURSE NAME: CSA07 COURSE CODE: Computer Networks

COURSE OUTCOME COVERED

CO1: Apply the functions of OSI layer in enabling reliable data transmission across networks. (BL2, BL3)

Assessment Tool Weightage: 30%

Annexure A

CONCEPT MAPPING

Code of Conduct:

I V. Jagadeesh (192312454) *(Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.*
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: V. Jagadeesh

Criteria	Weightage (%)	Q1 10marks	Q2 10marks	Q3 10marks
Concept Identification	20%	1.25	1.4	1.4
Concept Mapping Structure	25%	1.75	1.75	1.75
Linking & Relationships	20%	1.4	1.4	1.4
Application & Analysis	20%	1.4	1.4	1.4
Explanation & Clarity	15%	1.05	1.05	1.05

Total Marks 21 **/30**

Signature of the Faculty

CONCEPT MAPPING QUESTIONS

1. A partially completed concept map is given below:

Application → Presentation → _____ → Transport → Network → _____ → Physical.
Analyze the missing layers and explain how their functions are essential for reliable communication.

3. Construct a concept map linking data units (bits/nibble, packet, frame, byte) with respective OSI layers and analyze how transformation of data unit supports transmission.

4. Create the concept map:
Data link → MAC Address → Local delivery
Network → IP Address → Global Delivery
Explain how these relationships ensure complete data delivery across networks.

4. A concept map shows:
ErrorControl
→Data link layer →ErrorDetection
→ Transport Layer → Error Correction
Analyze how the interaction between these layers improves reliability in data transmission.

5. Develop a concept map for web browsing using OSI layers and analyze how failure in one layer affects the entire communication process.

6. A network issue is represented in the concept map:
Physical ✔ → Data Link ✔ → Network ✖ → Transport ✖ → Application ✖
Analyze the problem and explain how the concept map helps in troubleshooting.

RUBRICS FOR CALCULATION

Criteria	Weightage (%)	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Concept Identification	20%	Accurately identifies all relevant OSI layers, concepts, and relationships	Identifies most concepts with minor omissions	Identifies limited concepts, some inaccuracies	Unable to identify key concepts correctly
Concept Mapping Structure	25%	Constructs a clear, logical, and well-organized concept map with proper hierarchy and connections	Concept map is mostly clear with minor structural issues	Concept map lacks clarity, organization, or proper flow	No clear structure or incorrect mapping
Linking & Relationships	20%	Clearly establishes correct relationships between layers, functions, and processes	Relationships are mostly correct with minor errors	Limited or partially correct relationships	Incorrect or missing relationships

2. Construct a concept map linking data units (Segment, Packet, Frame, Bits) with their respective OSI layers and analyze how transformation of data units supports transmission.

3.Given the concept map:

DataLink→MACAddress→LocalDelivery

Network → IP Address → Global Delivery

Explain how these relationships ensure complete data delivery across networks.

4. A concept map shows:

ErrorControl

→DataLinkLayer→ErrorDetection

→ Transport Layer → Error Correction

Analyze how the interaction between these layers improves reliability in data transmission.

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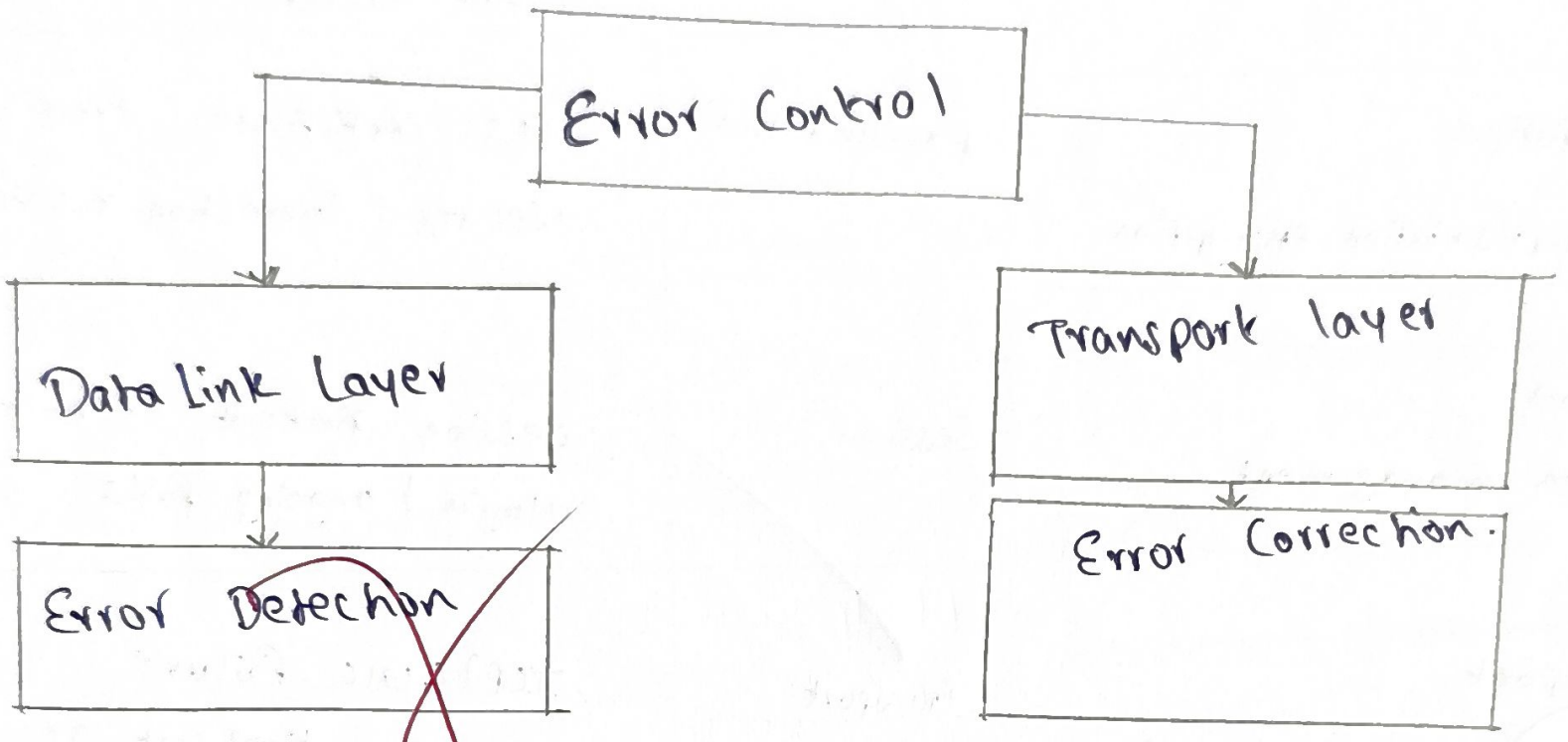
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CSA0710 - COMPUTER NETWORKS

Error Control in the Dark Link and Transport Layers.

Concept map.



Theory and Analysis:

1. Data Link layer - Error Detection

* Operates on a single link (hop-to-hop)
* Uses techniques like Cyclic Redudancy check (CRC) parity check, checksum.
* Detects errors in frames and discards corrupted frames.
* Some protocols can also request retransmission (eg. ARQ)

2. Transport layer - Error Correction

* Provides end-to-end reliability (eg. TCP)
* Uses sequence number, acknowledgements (ACKs) and retransmissions to recover lost or corrupted data.
* Ensures data is delivered correctly and in order to the application.

3. Web Browsing using OSI layers.



7. Application

HTTP/HTTPS . DNS . Browser request

6. Presentation

Data representation . encryption

5. Session

Session management

4. Transport

TCP or QUIC over UDP

3. Network

IP addressing & routing

2. Data link

frames & MAC addressing

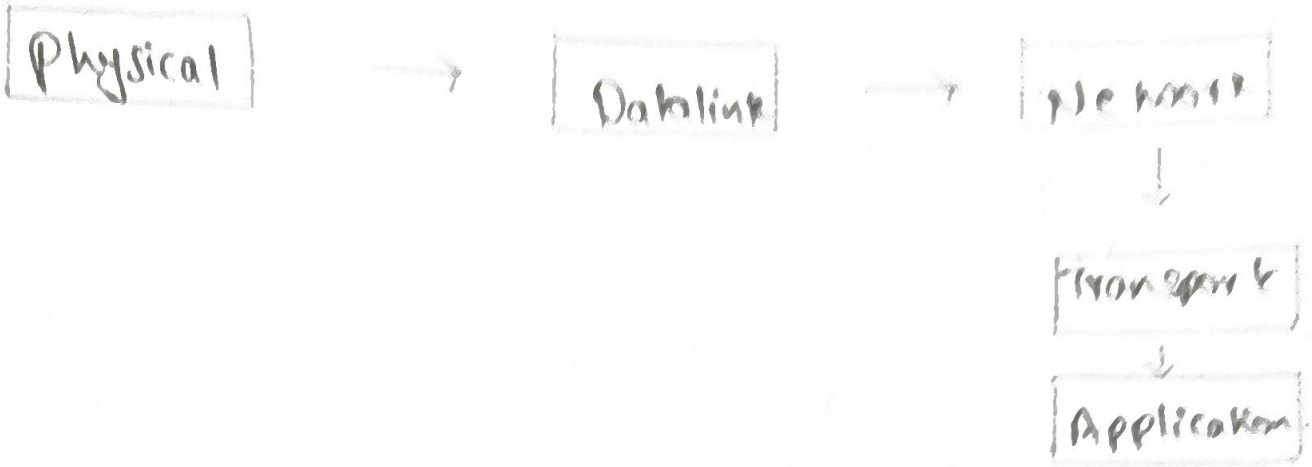
1. Physical

Electrical / Optical or radio signals

How failure in one layer affects the process

Layer	effect of failure
Application	Server → website not found / page not loaded.
Presentation	TLS certificate error → secure connection rejected
Session.	Session timeout or failure → login / ongoing fails
Transport	TCP / QUIC failure → connection time out or transfer stops
Network	incorrect IP route or no route to host → server cannot be reached.
Data link	wifi / ethernet failure → local network connectivity lost
Physical	Broken cable / weak signal. → bits cannot be transmitted reliability.

Troubleshooting a network using the concept map



Analysis of the problem.

* The network layer has failed, which means IP Addressing routing or Path to the destination is not working.

* Since the transport layer depends on the network layer, it also fails.

Possible Causes at the Network layer.

* Incorrect IP address or Subnet mask
* Missing or incorrect default gateway.
* Routing tables errors.
* Destination IP unreachable.